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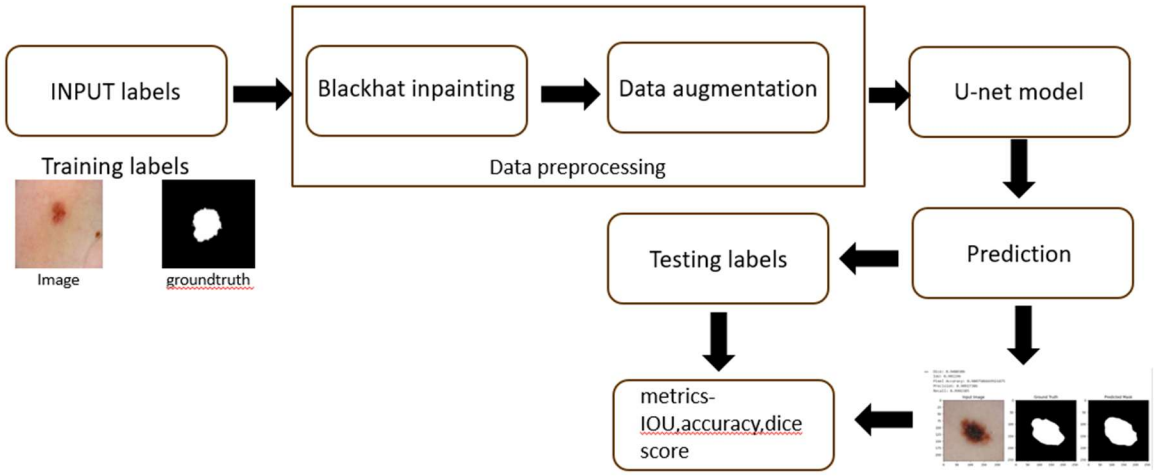
Project Report Group 10 : Automatic Skin Lesion Segmentation from Dermoscopic Images

1. Introduction

Skin cancer is one of the most common forms of cancer worldwide. Early detection plays a critical role in improving patient outcomes. Automated classification of skin lesions using machine learning provides a scalable and objective approach to assist diagnosis.

This project focuses on segmentation of skin cancer images and building ,evaluating multiple machine learning models to classify skin lesions as **Benign** or **Malignant** using extracted image features. The aim is to compare different models and identify the most effective approach based on performance metrics.

Workflow of this project:



Pipeline of this project:

1 Raw dermoscopic images are collected in a dataset folder.

2 Hair removal is performed using BlackHat morphological operation.

3 Detected hair pixels are removed using image inpainting.

4 Images are resized to a fixed resolution of 256 × 256 pixels.

5 Pixel values are normalized to the range [0,1].

6 The processed images are saved into a new folder for training and validation.Metrics such DICE score,IOU, pixel accuracy,precision and recall

7.Trained model is exported as weights in .pth file and used for prediction

8.TEsting with unseen images and image metrics were calculated

Segmentation part

dataset used:

Standard ISIC dataset with Training images = 900 (with their ground truth masks)

Testing images = 360

1. Raw Image Input

The preprocessing pipeline begins with raw dermoscopic images stored in a dataset directory. Each image contains a skin lesion along with possible artifacts such as hair strands, illumination variations, and background noise. These artifacts can negatively affect the segmentation model's performance if not removed.

2. Hair Removal using BlackHat Morphological Operation

Hair is one of the most common artifacts in dermoscopic images. To detect hair structures, a morphological BlackHat operation is applied. The BlackHat transformation highlights dark structures (such as hair) on a lighter background by computing the difference between the closing of the image and the original image.

3. Hair Removal using Image Inpainting -------> preprocess.py

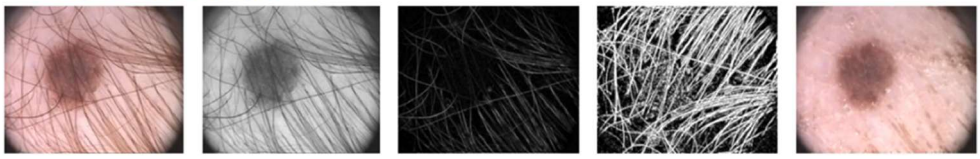
After detecting hair regions using the BlackHat operation, a binary mask is created using thresholding. This mask identifies pixels corresponding to hair. The OpenCV inpainting algorithm is then applied to fill these regions using neighboring pixel information. This effectively removes hair artifacts while preserving surrounding image structures.

BlackHat Transform:

$$BlackHat(I) = Closing(I) - I$$

Where

- I = original image
- Closing = dilation followed by erosion



(a) Original Image (b) Grayscale Image (c) Blackhat Filtered Image (d) Threshold for Inpainting (e) Final Inpainted Image

4. Image Resizing

Deep learning models require fixed-size inputs. Therefore, each image is resized to 256 × 256 pixels. This size is commonly used in convolutional neural networks and provides a balance between computational efficiency and feature preservation.

5. Image Normalization

Pixel intensities in the image are normalized by dividing each pixel value by 255. This scales pixel values from the range [0,255] to [0,1]. Normalization stabilizes the training process and helps neural networks converge faster.

6. Saving Processed Images

After preprocessing, the cleaned and resized images are saved into a new folder. The filenames remain the same as the original images to maintain consistency with the corresponding ground truth segmentation masks. These processed images are later used as input for the U-Net segmentation model.

Model training configuration:

The model that was used for segmentation is U-net model with 4 level architecture and the model details as follows:

Parameter	Value	Rationale
Framework	PyTorch / TensorFlow + Keras	Flexible, GPU-accelerated deep learning
Model	U-Net (custom implementation)	Proven SOTA for biomedical segmentation
Input Size	256 × 256 × 3	Balance between detail and memory
Batch Size	16	Stable gradient estimates
Epochs	30	Avoids overfitting
Optimizer	Adam (lr = 1e-4)	Adaptive, fast convergence
Loss	BCE + Dice Loss (0.5 + 0.5)	Handles class imbalance
Train/Val Split	80% / 20%	Standard split for generalization check
Augmentation	Flip (H/V), Rotation ±30°	Expanded effective training set

Training results:

The training results at epoch 30 Is convereged to be in dice loss =0.1188

```
Epoch: 1 Loss: 0.4104863660203086
Epoch: 2 Loss: 0.307211147248745
Epoch: 3 Loss: 0.26830449567900766
Epoch: 4 Loss: 0.2527064484026697
Epoch: 5 Loss: 0.2257139296995269
Epoch: 6 Loss: 0.21614140305254195
Epoch: 7 Loss: 0.2030058690243297
Epoch: 8 Loss: 0.18499038815498353
Epoch: 9 Loss: 0.18243562016222212
Epoch: 10 Loss: 0.17100368870629204
Epoch: 11 Loss: 0.16623833262258106
Epoch: 12 Loss: 0.17209456952081786
Epoch: 13 Loss: 0.16169099153743385
Epoch: 14 Loss: 0.15633132192823623
Epoch: 15 Loss: 0.1559318749441041
Epoch: 16 Loss: 0.15000716828637653
Epoch: 17 Loss: 0.1442921332187123
Epoch: 18 Loss: 0.1438463605940342
Epoch: 19 Loss: 0.13736685179173946
Epoch: 20 Loss: 0.13735328473978572
Epoch: 21 Loss: 0.1383975659807523
Epoch: 22 Loss: 0.13664996268020735
Epoch: 23 Loss: 0.13221249373422728
Epoch: 24 Loss: 0.13559137392375203
Epoch: 25 Loss: 0.12476483682791392
Epoch: 26 Loss: 0.1247853713730971
Epoch: 27 Loss: 0.12986319458319082
Epoch: 28 Loss: 0.13981937799188826
Epoch: 29 Loss: 0.12298267839683427
Epoch: 30 Loss: 0.11888299410541853
```

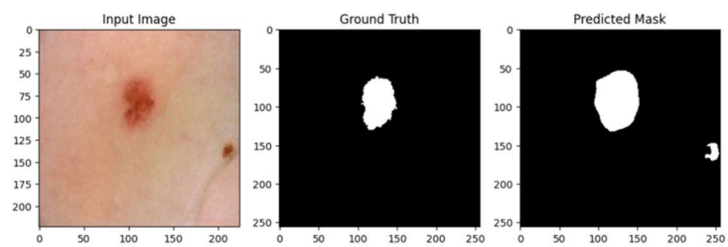
Image metrics

```
Dice: tensor(0.8442, device='cuda:0')
IoU: tensor(0.7337, device='cuda:0')
Pixel Accuracy: tensor(0.9419, device='cuda:0')
Precision: tensor(0.8848, device='cuda:0')
Recall: tensor(0.8962, device='cuda:0')
```

Training loss after 30 epochs = 0.1188

Validation results:

```
Dice: 0.94141084
IoU: 0.88930714
Pixel Accuracy: 0.9422454833984375
Precision: 1.0
Recall: 0.889307
```



The following metrics were obtained on the testing dataset, which was not used during training.

Metric Value
Dice Score 0.9414
IoU 0.8893
Pixel Accuracy 0.9422
Precision 1.0000
Recall 0.8893
Later, the model weights was saved into unet_segmentation.pth file for pediction

Data analysis

2. Dataset Description

The dataset used in this project consists of pre-extracted 360 benign and 300 malignant dermoscopic images. For each of them ABCD features is extracted and saved in .Xlsx files.Each row corresponds to one image.

Handcrafted features extraction:

Features used:

- **Asymmetry** – Measures irregularity in lesion shape
- **Border Compactness** – Indicates boundary smoothness
- **Color Variance** – Variation in color distribution
- **Color Entropy** – Complexity of color patterns
- **Diameter (px)** – Size of the lesion
- **Area Ratio** – Relative area coverage

Target Variable:

- **Cancer_type**
    - Benign
    - Malignant

Data Cleaning Steps:

- Removed non-relevant classes (only kept Benign and Malignant)
- Converted all feature columns to numeric
- Removed missing values

4. Data Preprocessing

4.1 Label Encoding

The target variable was converted:

- Benign → 0
- Malignant → 1

4.2 Feature Scaling

Standardization was applied using **StandardScaler**:

- Mean = 0
- Standard deviation = 1

This step is important for models like SVM and Logistic Regression.

4.3 Train-Test Split

The dataset was split into:

- **Training set: 80%**
- **Validation set: 20%**

Stratified sampling was used to maintain class balance.

5. Machine Learning Models

The following models were implemented:

1. Logistic Regression

- Linear model for binary classification
- Key parameter: max_iter = 1000

2. Decision Tree

- Rule-based model
- Key parameter: max_depth = 10

3. Random Forest

- Ensemble of decision trees
- Key parameter: n_estimators = 100

4. Support Vector Machine (SVM)

- Finds optimal separating hyperplane
- Uses probability estimation (probability=True)

6. Model Evaluation Metrics

The following metrics were used:

- **Accuracy** – Overall correctness
- **Precision** – Correct positive predictions
- **Recall** – Ability to detect malignant cases
- **F1 Score** – Balance between precision and recall
- **AUC (ROC Curve)** – Model discrimination ability

7. Results and Comparison

A comparison table was generated for all models and saved as:

- model_comparison.csv
- model_comparison.png

Model	Accuracy	Precision	Recall	F1 Score	AUC
Logistic Regression	0.705	0.744	0.533	0.621	0.794
Decision Tree	0.606	0.571	0.533	0.552	0.62
Random Forest	0.674	0.681	0.533	0.598	0.759
SVM	0.705	0.744	0.533	0.621	0.798

Interpretation:

- Models were compared across multiple metrics rather than just accuracy
- AUC and F1 score were particularly important due to medical context
- Recall is critical because missing malignant cases is risky

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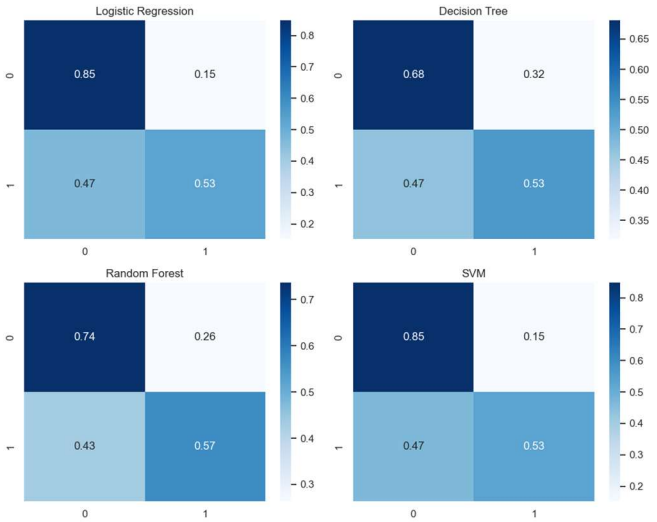
8. Confusion Matrix Analysis

Confusion matrices were plotted for all models.



Each matrix shows:

- True Positives
- True Negatives
- False Positives
- False Negatives

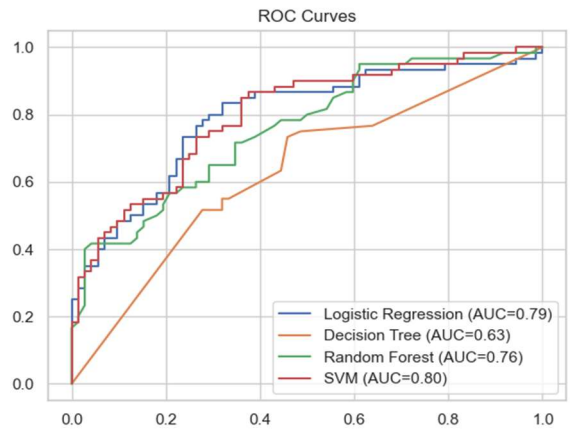
Values were normalized to better compare performance.

Insight:

- Helps identify whether the model is biased toward a class
- Especially useful for understanding false negatives (critical in cancer detection)

9. ROC Curve Analysis

ROC curves were plotted for all models.



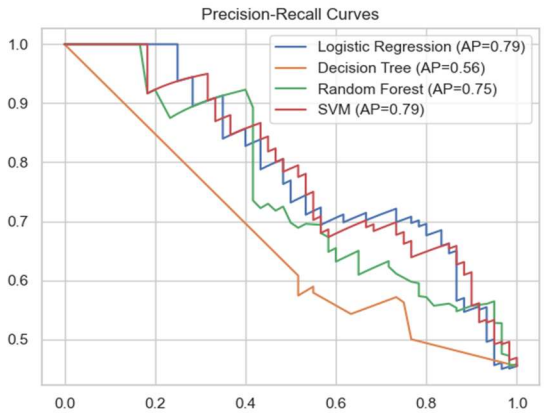
- X-axis: False Positive Rate
- Y-axis: True Positive Rate

AUC Interpretation:

- Closer to 1 → better model
- Around 0.5 → random guessing

10. Precision-Recall Curve

Precision-Recall curves were also generated.



Why important here:

- Better suited for imbalanced datasets
- Focuses on positive class (Malignant)

11. Feature Importance

Feature importance was computed using the Random Forest model.

This identifies which features contribute most to predictions.

Usefulness:

- Provides interpretability
- Helps validate biological relevance of features

12. Decision Tree Visualization

The trained decision tree was visualized.

Purpose:

- Understand decision rules
- Interpret model behavior step-by-step

13. Test Set Evaluation

An independent test dataset was used (if provided).

- Same preprocessing steps applied
- Accuracy calculated for each model

This ensures that the model generalizes beyond training data.

14. Discussion

- Multiple models were tested to ensure robustness
- Ensemble methods (Random Forest) typically perform better due to reduced variance
- SVM performs well in high-dimensional spaces
- Logistic Regression provides interpretability
- Decision Trees offer clear rule-based insights

In a medical setting:

- **Recall is more important than accuracy**
- False negatives (missed cancer cases) must be minimized

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15. Conclusion

This project successfully implemented a complete segmentation using Unet and machine learning pipeline for skin lesion classification.

Key takeaways:

- Proper preprocessing and scaling significantly improve performance
- Ensemble models provide strong results
- Model evaluation must consider multiple metrics
- Visualization aids interpretation and validation